

Dự đoán nguy cơ đánh giá xấu của đơn hàng Olist

Early-warning binary classification model using operational, payment, product, and customer features.

96,361

delivered orders after
joins

21.07%

bad-review target rate

4

models compared



Logistic Regression

Chosen by Negative F1, not raw
accuracy

The useful prediction is not “who is happy”; it is “who may complain.”

Olist needs a warning at delivery time, before the customer leaves a low review.

Target variable

- Negative = review score 1–3
- Positive = review score 4–5
- Business priority: detect Negative

Prediction moment

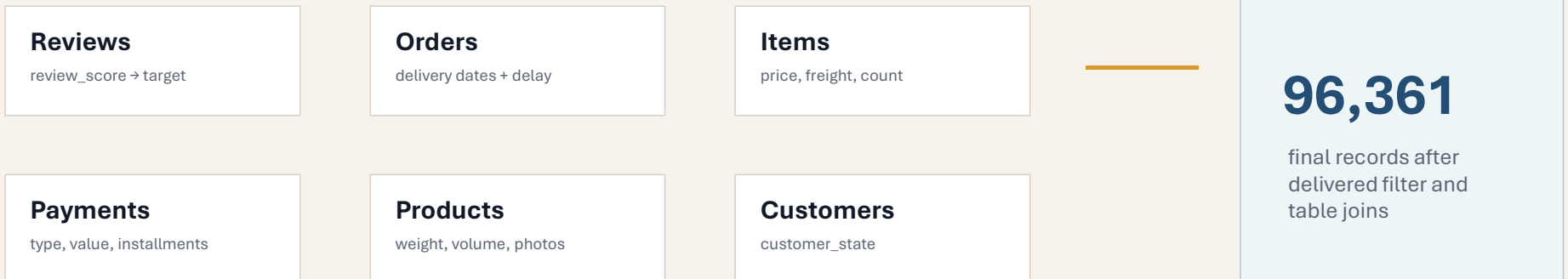
- Order status is delivered
- Use only known order history
- Trigger support action if risk is high

Why metric choice matters

- Accuracy rewards the majority class
- Negative is only 21.07% of data
- Negative F1 is better aligned

Six Olist tables become one modeling dataset with 96,361 delivered orders.

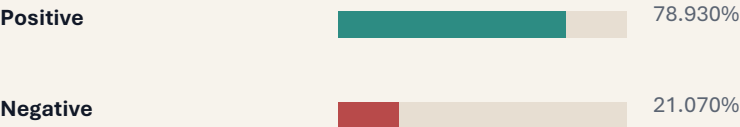
The join keeps the model focused on data available around the delivery event.



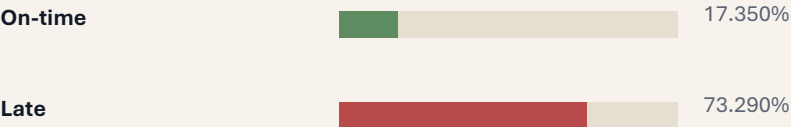
Bad reviews are a minority overall, but late delivery flips the risk profile.

The strongest simple signal is operational: late orders are far more likely to be Negative.

Class distribution



Negative review rate by delivery status



9 days

median delivery time for Positive orders

13 days

median delivery time for Negative orders

6.65%

orders delivered late

The model combines operational friction, cost pressure, product complexity, and customer context.

Feature engineering turns raw timestamps and transaction fields into predictive signals.

Operational

- delivery_days
- estimated_delivery_days
- delay_days
- is_late

Cost & payment

- total_price
- total_freight
- freight_ratio
- payment_type

Product

- product_weight_g
- product_volume_cm3
- photos
- description length

Customer

- customer_state
- installments
- item_count

Identifiers and raw review score are excluded from model inputs to avoid leakage.

Four classifiers were trained under the same preprocessing pipeline.

Numerical features are imputed and scaled; categorical features are imputed and one-hot encoded.

01

Logistic Regression

linear baseline + class_weight

02

Decision Tree

interpretable non-linear splits

03

Random Forest

bagged tree ensemble

04

Gradient Boosting

sequential error-correcting trees

80/20

stratified train/test split

Negative F1

final selection metric

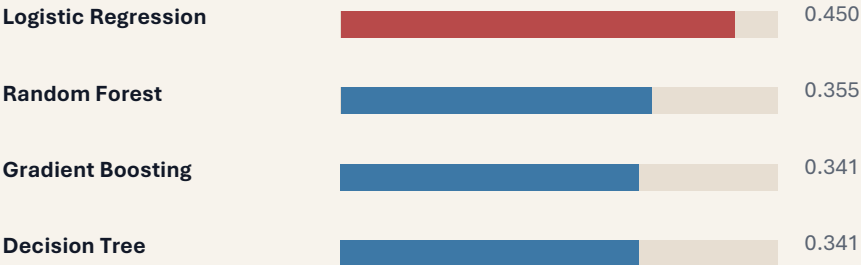
ROC-AUC

ranking quality support metric

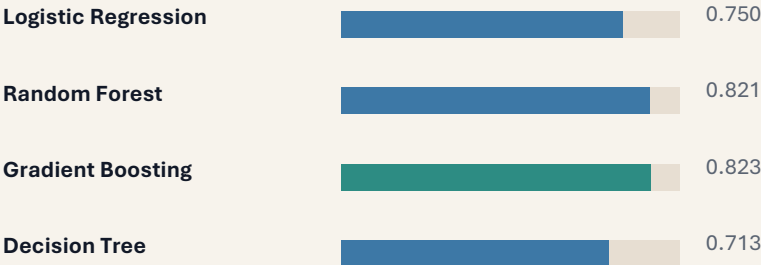
Gradient Boosting wins on accuracy; Logistic Regression wins on bad-review detection.

This tradeoff matters because the project is an early-warning system.

Negative F1: warning-objective metric



Accuracy: majority-class-friendly metric



Decision: use Logistic Regression as the recommended baseline because it catches more Negative cases.

Logistic Regression is the better baseline for customer-care intervention.

It sacrifices some accuracy to improve detection of the minority Negative class.

	Predicted Negative	Predicted Positive
Actual Negative	1972	2090
Actual Positive	2723	12488

0.450

Negative F1

0.485

Negative Recall

0.701

ROC-AUC

Late delivery is the clearest risk signal, and threshold tuning is the next lever.

The current model is a solid baseline, but the warning system should be optimized for recall on bad reviews.

- Operational delay has a much stronger relationship with Negative reviews than payment type alone.
- Accuracy is misleading because Positive orders dominate the dataset.
- Use Logistic Regression as the first deployable baseline for explainability and Negative-class detection.
- Tune the decision threshold to catch more high-risk customers before support outreach.

Recommended next step: optimize for business cost, not just statistical score.

THANK YOU